



PROBLEM STATEMENT ID- 25071

**PROBLEM STATEMENT TITLE- AI-BASED ROCKFALL PREDICTION
AND ALERT SYSTEM FOR OPEN-PIT MINES**

THEME- DISASTER MANAGEMENT

PS CATEGORY- SOFTWARE

TEAM NAME- FALLEN ROCK

TEAM ID- 85618

MEDIA ARCHIVE



GITHUB-REPO



DASHBOARD DEMO



REPORT



WEBSITE



Proposed Solution: ML Model & 3D Hazard Rendering



Step 1 → Raw Simulation Data

Rocfall3 outputs:

Seeder coordinates (x, y, z)
Slope height & roughness
Vibration, pore pressure
Events (collisions, bounces)

Step 2 → Feature Engineering

Derived attributes:

Slope angle
Normalized coordinates
Log-transformed travel time
Event density

Step 3 → Hazard Labeling

Rule: Runout > 150 m = Hazard (Red)
Runout ≤ 150 m = Safe (Blue)

Step 5 → 3D Visualization & Dashboard

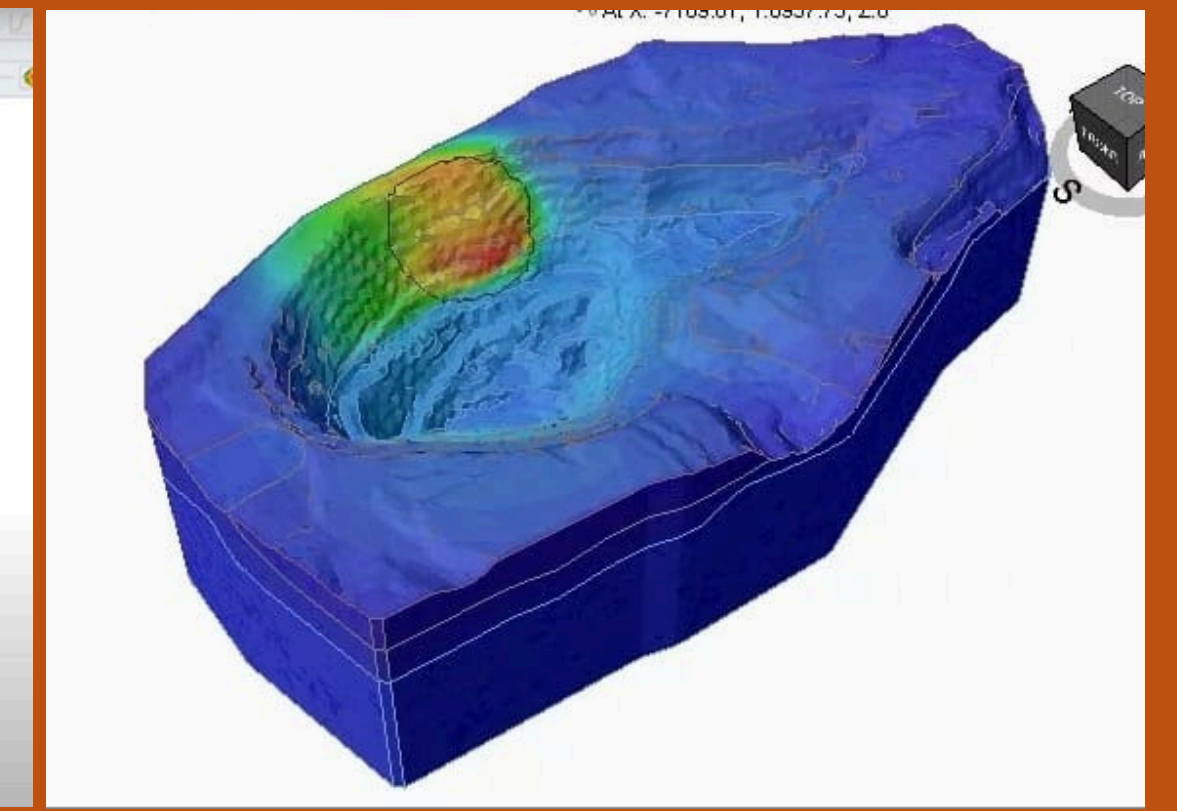
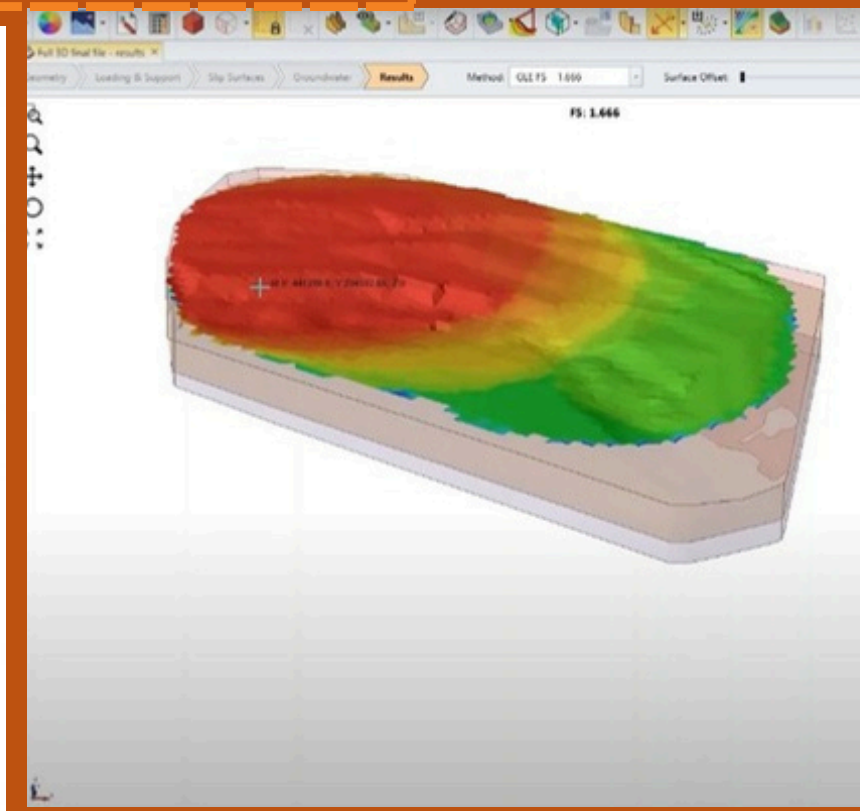
Hazard probability mapped onto slope
Red = high risk, Blue = safe
Interactive dashboard + QR codes for dataset & results

Step 4 → Machine Learning

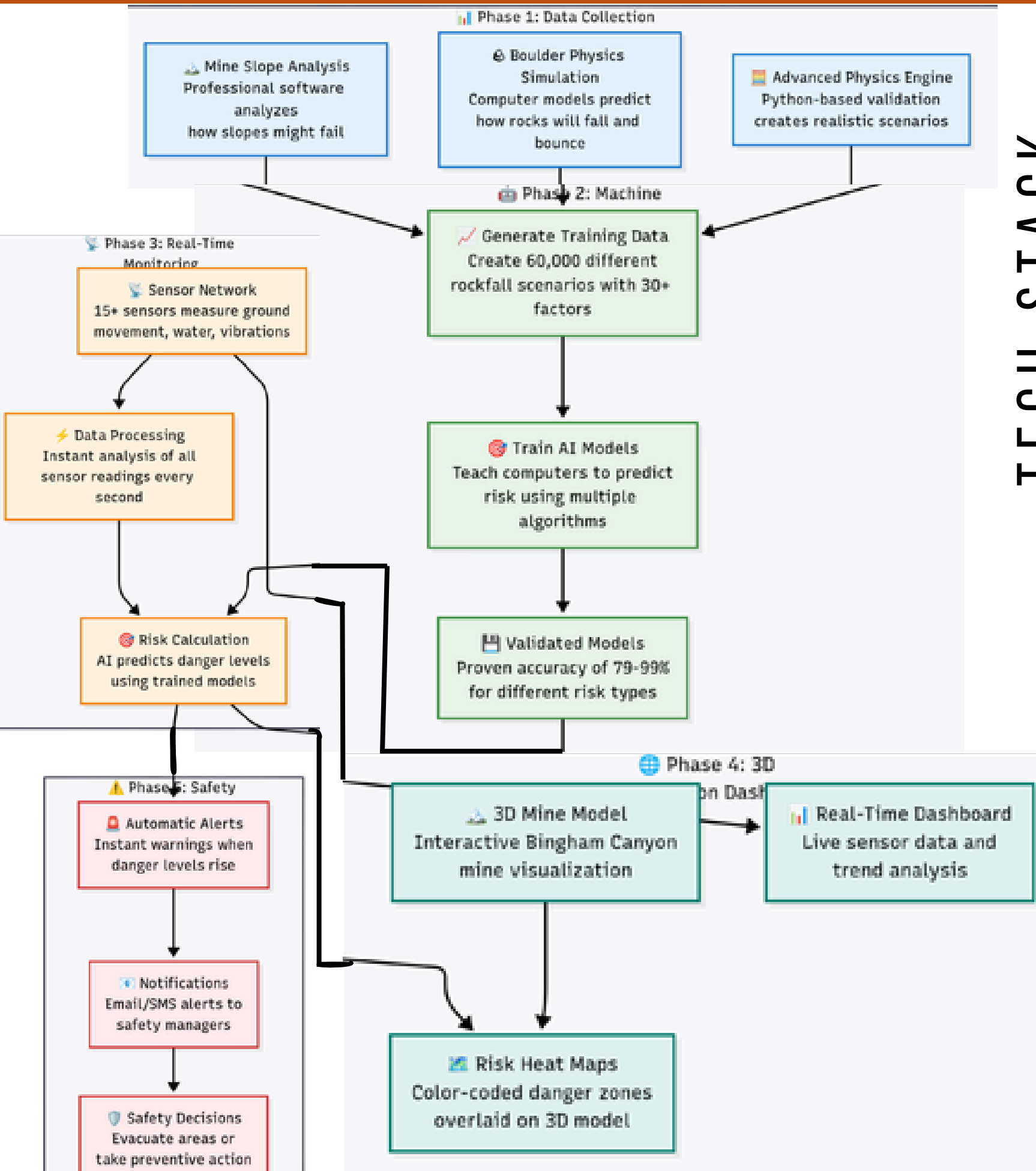
Model (XGBoost)
Accuracy: ~ 84%
ROC-AUC: ~ 0.87
Top Predictors: slope, seeder elevation (z), travel time, rainfall

Dataset snapshot few columns:

Slope Angle (°)	Seeder Height (m)	Local Relief (m)	Rainfall 7d (mm)	API 7d	Disp. Rate (mm/day)	Risk Probability (7d)	Risk Severity
43.1	36.6	43.8	17.7	17.4	0.34	0.27	0.35
50.6	39.8	23.7	22.6	18.0	1.18	0.28	0.19
70.5	46.0	52.8	13.2	10.1	0.94	0.28	0.32
57.3	44.6	7.2	35.9	1.05	59.6	0.32	0.40



Technology Employed



TECH STACK

WEB DEVELOPMENT

NEXT.JS, TYPESCRIPT, HTML, CSS , TAILWIND,
REACT.JS, POSTGRESQL, NODE.JS

DATA FORMATS

CSV, NPZ, GLB/GLTF, GIF

IMAGE PROCESSING

TRIMESH, THREE.JS

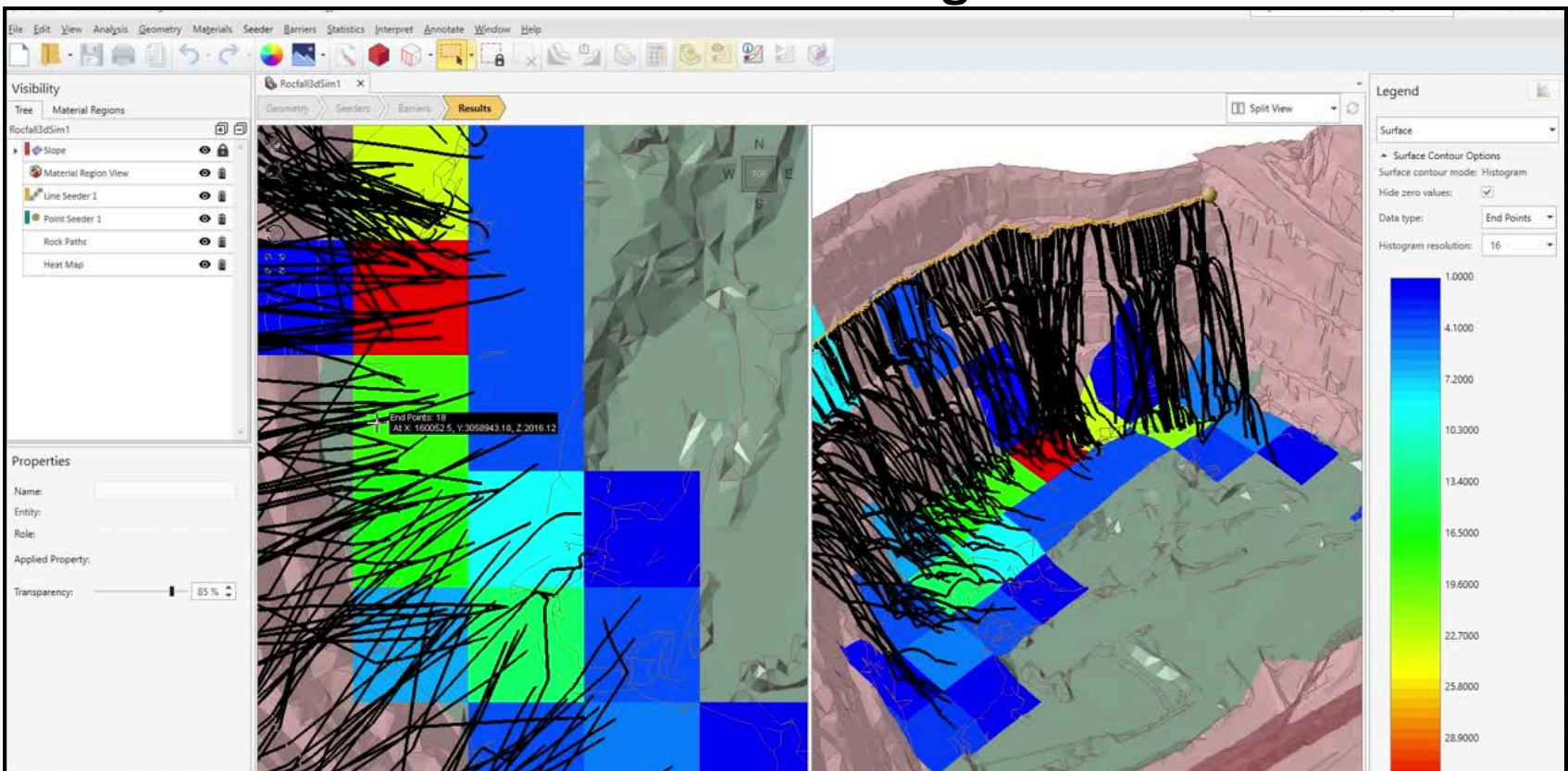
MODEL
DEVELOPMENT

NUMPY, PANDAS, SCIKIT-LEARN, SCI-PY, JOBLIB, MATPLOTLIB,
PLOTLY, SEABORN

HOSTING AND
DEPLOYMENT

VERCEL

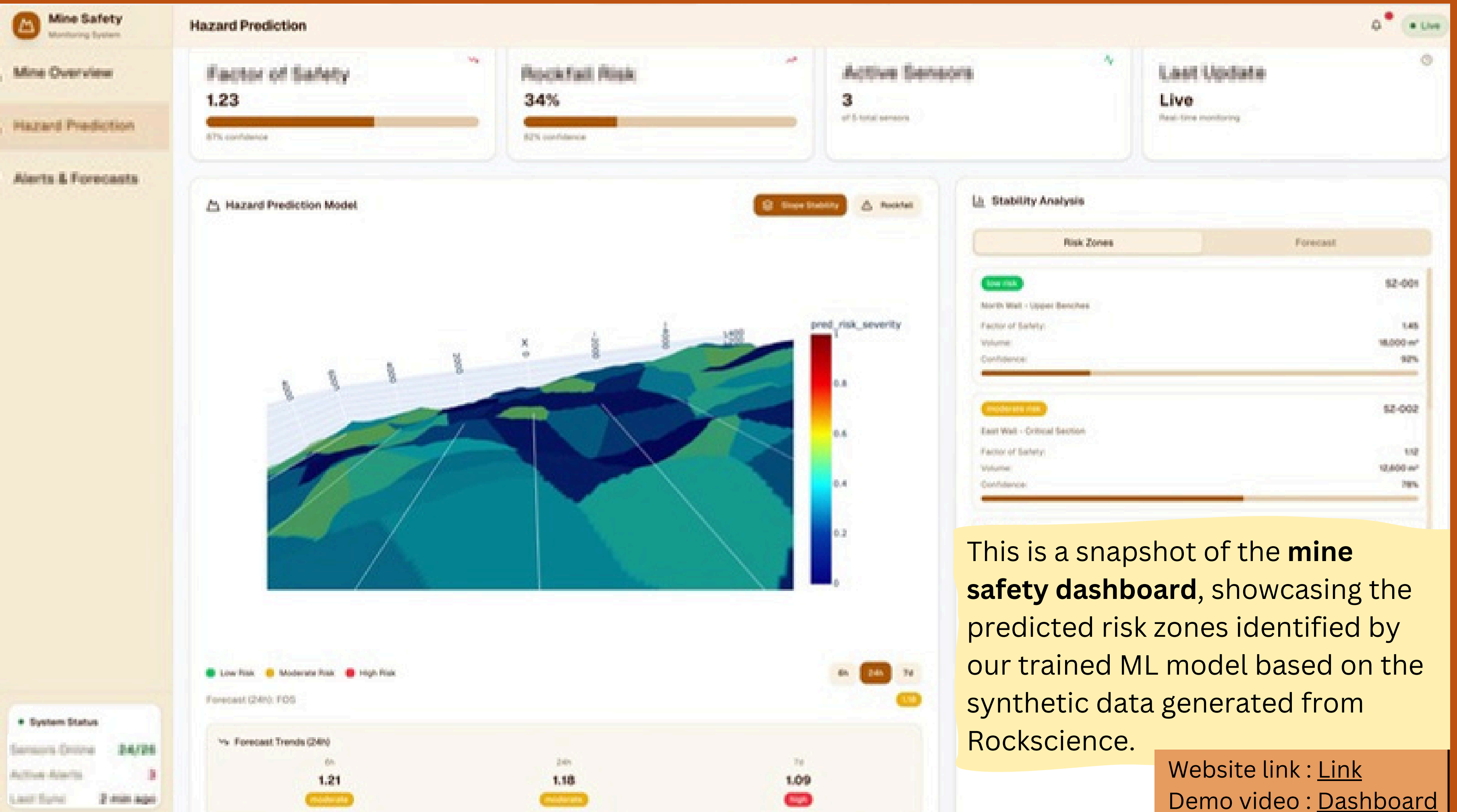
Simulation in Rocscience showing Rockfall of seeders



Mining Safety Monitoring Dashboard



SMART INDIA HACKATHON

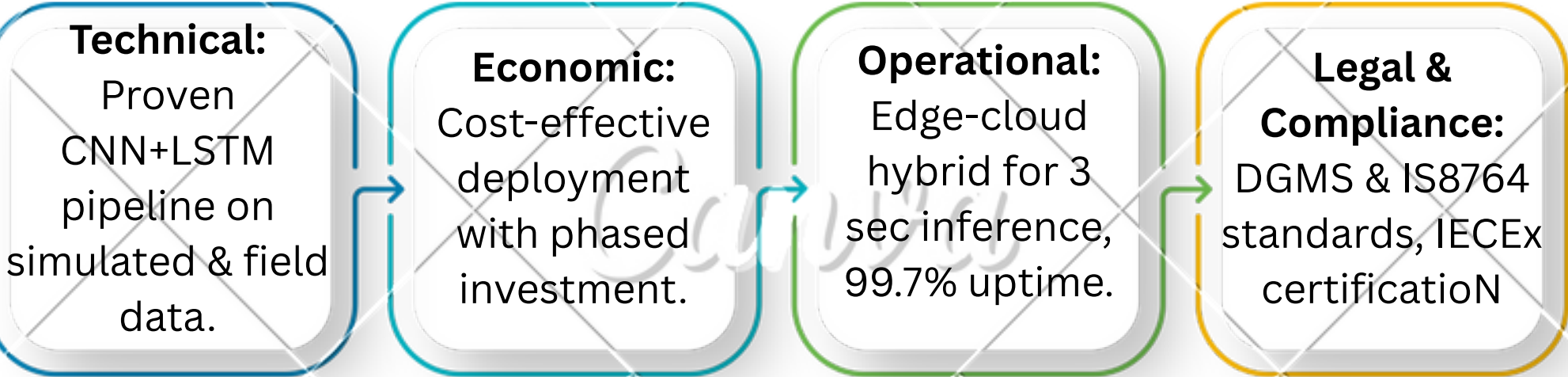


This is a snapshot of the **mine safety dashboard**, showcasing the predicted risk zones identified by our trained ML model based on the synthetic data generated from Rockscience.

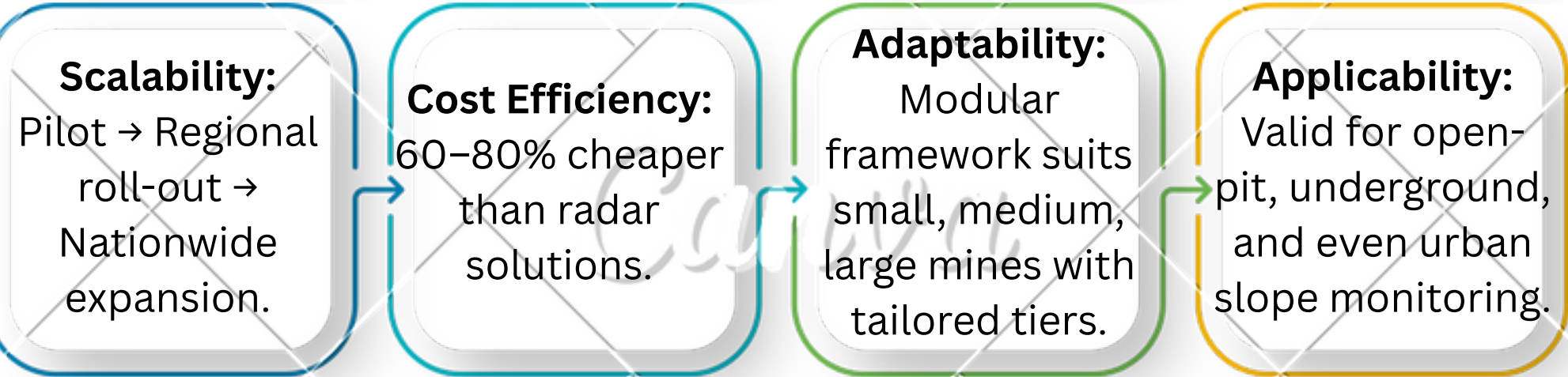
Website link : [Link](#)
Demo video : [Dashboard](#)

Feasibility and Viability:

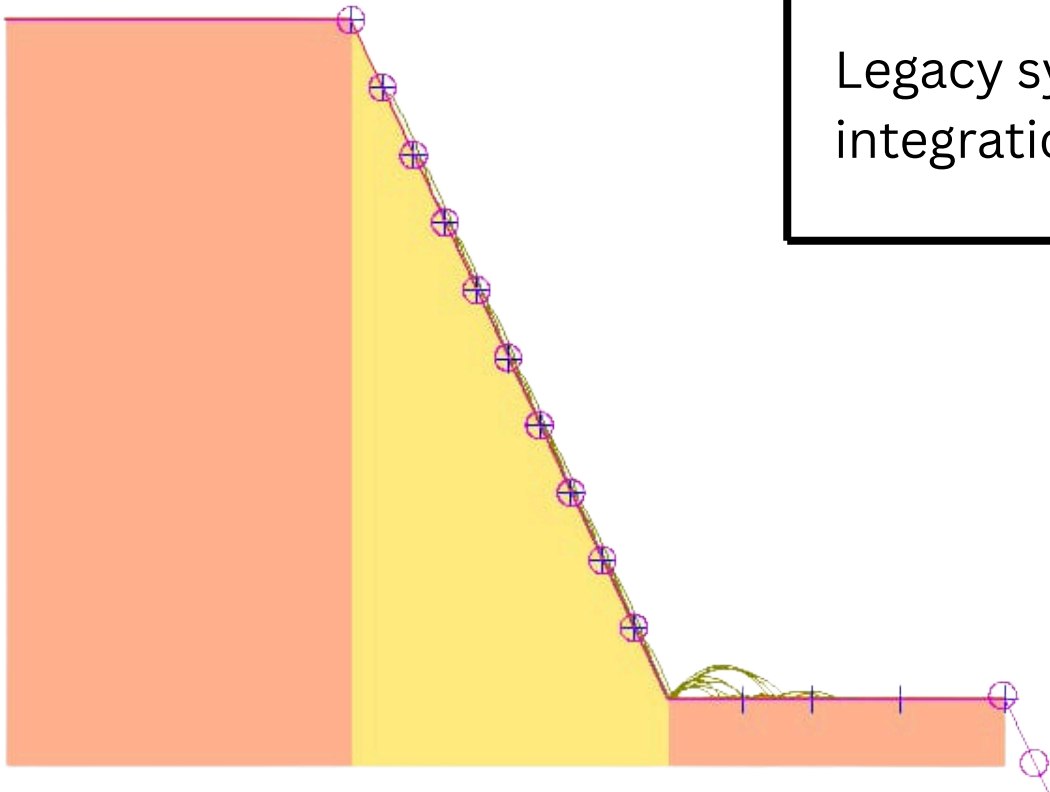
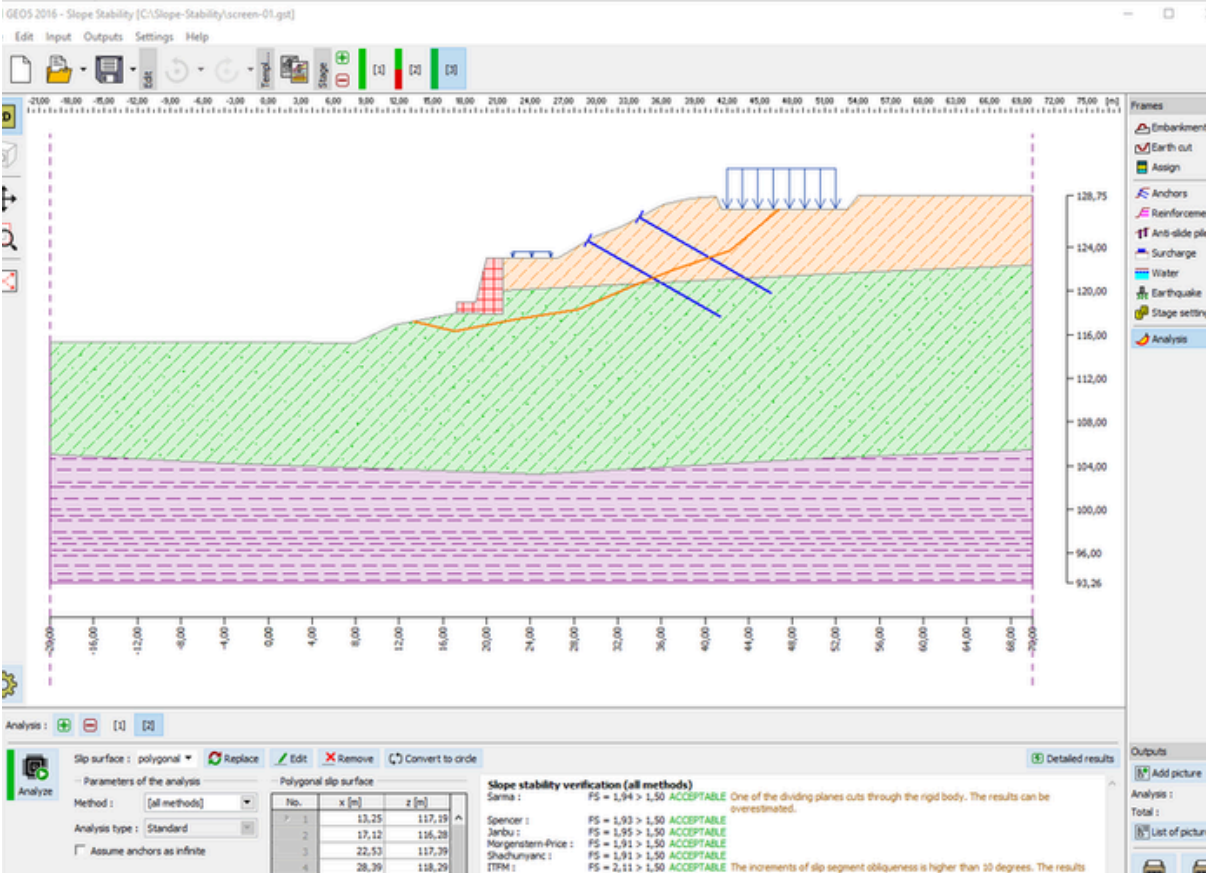
Feasibility:



Viability:



RISK	MITIGATION
Sensor loss from rock impact	Impact resistant casings, Strategic positioning, Redundant coverage
Connectivity Outrages & Extreme weather	Edge computing fallback, multi-network (4G/5G/satellite) switching; 48-hr data cache
Model Overfitting	Transfer learning, Rigorous cross-validation, L2/L1 regularization
Budget Overruns	Real-time cost monitoring dashboard, phased deployment
Legacy system integration	Standardized APIs, modular adapters, phased integration roadmap



The images on the left showcase the 2D simulations created in Rockscience Rockfall 2D and Rockslope 2D



Impact and Benefits

Metric	Historic Loss (2000–2025)	Projected with AI Soln
Economic Loss	\$8-10 billion	60–90% reduction
Fatalities	45,000+	80% fewer deaths
Major Equipment Lost	\$2–5M per incident	Largely preventable
Affected Land (one mine)	75 km ²	Significantly less
Social Displacement	Severe, long-term	Greatly diminished

Existing Solutions

1.Slope Stability Monitoring

- Deployment of Slope Stability Radar (SSR) systems for real time monitoring with sub-millimeter precision detection.
- Initially mandated for top 35 coal mines; currently only one active in Hindustan Zinc, Rajasthan.
- Phase-wise deployment plan spanning 2022-2030.

2.Systematic Slope Monitoring

- Scientific slope monitoring replacing visual inspection only, using instruments like inclinometers, piezometers, extensometers, total stations.
- Maintaining Slope Monitoring Programme (SMP) document and establishment of Trigger Action Response Plans (TARP) for slope failures.

References:

Research Papers:

- Senanayake, I. P., Hartmann, P., Giacomini, A., Huang, J., & Thoeni, K. (2024). Prediction of rockfall hazard in open pit mines using a regression based machine learning model. *International Journal of Rock Mechanics and Mining Sciences*, 177, 105727. | [Link](#)
- Farmakis, I., DiFrancesco, P. M., Hutchinson, D. J., & Vlachopoulos, N. (2022). Rockfall detection using LiDAR and deep learning. *Engineering Geology*, 309, 106836. | [Link](#)

[Report link](#) : [link](#)

